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RESEAU, SECURITE ET SYSTEMES DISTRIBUES(R2SD)

Activité 2 : Installation de Nagios

Sous la direction

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C'est quoi Nagios ?

Nagios est un logiciel de surveillance réseau open source largement utilisé pour surveiller l'état de divers composants d'un réseau informatique. Il permet aux administrateurs système de surveiller en temps réel divers paramètres, tels que la disponibilité des serveurs, la performance du réseau, la charge CPU, l'utilisation de la mémoire, la consommation de la bande passante. L'objectif principal de Nagios est d'aider les équipes informatiques à détecter rapidement les problèmes potentiels et à résoudre les pannes avant qu'elles n'affectent sérieusement les opérations commerciales.

Nagios fonctionne en interrogeant périodiquement des hôtes et des services sur le réseau à l'aide de sondes appelées « plugins ». Ces plugins sont des scripts ou des programmes qui exécutent des vérifications spécifiques sur des systèmes distants. Si un problème est détecté, Nagios peut avertir les administrateurs par le biais de notifications, telles que des courriels, des SMS ou des alertes sur un tableau de bord.

I. Installation de Nagios

Nous ouvrons d'abord le terminal Ubuntu et mettons à jour le système :

sudo apt update

Maintenant, nous installons les dépendances nécessaires :

```
admin-sys@UbuntuServer:~$ sudo apt install autoconf bc gawk dc build-essential gcc libc6 make wget unzip apache2 php
libapache2-mod-php libgd-dev libmcrypt-dev make libssl-dev snmp libnet-snmp-perl gettext
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
bc is already the newest version (1.07.1-3build1).
bc set to manually installed.
dc is already the newest version (1.07.1-3build1).
dc set to manually installed.
php is already the newest version (2:8.1+92ubuntu1).
wget is already the newest version (1.21.2-2ubuntu1).
wget set to manually installed.
apache2 is already the newest version (2.4.52-1ubuntu4.6).
libc6 is already the newest version (2.35-0ubuntu3.4).
libc6 set to manually installed.
snmp is already the newest version (5.9.1+dfsg-1ubuntu2.6).
unzip is already the newest version (6.0-26ubuntu3.1).
unzip set to manually installed.
The following packages were automatically installed and are no longer required:
  libflashrom1 libftdi1-2 liblvm13 linux-headers-6.2.0-26-generic linux-hwe-6.2-headers-6.2.0-26
  linux-image-6.2.0-26-generic linux-modules-6.2.0-26-generic linux-modules-extra-6.2.0-26-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  automake autotools-dev binutils binutils-common binutils-x86-64-linux-gnu dpkg-dev fakeroot g++ g++-11 gcc-11
  libalgorithm-diff-perl libalgorithm-diff-xs-perl libalgorithm-merge-perl libasan6 libbinutils libbrotli-dev
  libc-dev-bin libc-devtools libc6-dev libcc1-0 libcrypt-dev libctf-nobfd0 libctf0 libdeflate-dev libdpkg-perl
  libexpat1-dev libfakeroot libfile-fcntllock-perl libfontconfig-dev libfreetype-dev libfreetype6-dev libgcc-11-dev
  libice-dev libitm1 libjbig-dev libjpeg-dev libjpeg-turbo8-dev libjpeg8-dev liblsan0 liblzma-dev libmcrypt4
  libnsl-dev libpng-dev libpng-tools libpthread-stubs0-dev libquadmath0 libsigsegv2 libsm-dev libstdc++-11-dev
  libtiff-dev libtiffxx5 libtirpc-dev libubsan0 libubsan1 libvpx-dev libx11-dev libxau-dev libxcb1-dev libxdmcp-dev
  libxpm-dev libxt-dev linux-libc-dev lto-disabled-list m4 manpages-dev pkg-config rpcsvc-proto uuid-dev
  x11proto-dev xorg-sgml-doctools xtrans-dev zlib1g-dev
```

Maintenant, le processus de téléchargement et d'installation de ces dépendances sont bien réalisés.

Nous allons télécharger la dernière version de nagios.

Nous extrayons le contenu du fichier Nagios téléchargé :

```

admin-sys@UbuntuServer:~$ cd Downloads/
admin-sys@UbuntuServer:~/Downloads$ ls
glpi-10.0.9.tgz  nagios-4.4.6.tar.gz
admin-sys@UbuntuServer:~/Downloads$ tar -xvf nagios-4.4.6.tar.gz
admin-sys@UbuntuServer:~/Downloads$ ls
glpi-10.0.9.tgz  nagios-4.4.6  nagios-4.4.6.tar.gz
admin-sys@UbuntuServer:~/Downloads$ cd nagios-4.4.6/
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$ ls
aclocal.m4      config.guess      docs              indent.sh         Makefile.in       pkginfo.in        tap              worker
autoconf-macos config.sub         doxy.conf         INSTALLING        make-tarball      README.md         test            xdata
base            configure         functions         install-sh        mkpackage         sample-config     THANKS
cgi             configure.ac      html             LEGAL             module            startup           t-tap
Changelog       contrib           include           lib               nagios.spec       subst.in          update-version
common          CONTRIBUTING.md   indent-all.sh   LICENSE           nagios.sysconfig  t                UPGRADING
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$

```

Nous compilons Nagios en exécutant :

```

admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$ sudo ./configure --with-httpd-conf=/etc/apache2/sites-enabled
checking for a BSD-compatible install... /usr/bin/install -c
checking build system type... x86_64-pc-linux-gnu
checking host system type... x86_64-pc-linux-gnu
checking for gcc... gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
checking whether we are cross compiling... no
checking for suffix of object files... o
checking whether we are using the GNU C compiler... yes
checking whether gcc accepts -g... yes
checking for gcc option to accept ISO C89... none needed
checking whether make sets $(MAKE)... yes
checking whether ln -s works... yes
checking for strip... /usr/bin/strip
checking how to run the C preprocessor... gcc -E
checking for grep that handles long lines and -e... /usr/bin/grep
checking for egrep... /usr/bin/grep -E
checking for ANSI C header files... yes
checking whether time.h and sys/time.h may both be included... yes
checking for sys/wait.h that is POSIX.1 compatible... yes
checking for sys/types.h... yes
checking for sys/stat.h... yes
checking for stdlib.h... yes
checking for string.h... yes
checking for memory.h... yes
checking for strings.h... yes
checking for inttypes.h... yes
checking for stdint.h... yes
checking forunistd.h... yes
checking forarpa/inet.h usability... yes

```

Nous exécutons la commande suivante pour compiler le programme Nagios :

```

admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$ sudo make all
cd ./base && make
make[1]: Entering directory '/home/admin-sys/Downloads/nagios-4.4.6/base'
gcc -Wall -I.. -g -O2 -DHAVE_CONFIG_H -DNSCORE -c -o nagios.o nagios.c
nagios.c: In function 'main':
nagios.c:611:25: warning: ignoring return value of 'asprintf' declared with attribute 'warn_unused_result' [-Wunused-result]
   611 |                 asprintf(&mac->x[MACRO_PROCESSSTARTTIME], "%llu", (unsigned long long)program_start);
       |                 ^~~~~~
nagios.c:841:25: warning: ignoring return value of 'asprintf' declared with attribute 'warn_unused_result' [-Wunused-result]
   841 |                 asprintf(&mac->x[MACRO_EVENTSTARTTIME], "%llu", (unsigned long long)event_start);
       |                 ^~~~~~
nagios.c: In function 'nagios_core_worker':
nagios.c:176:17: warning: ignoring return value of 'read' declared with attribute 'warn_unused_result' [-Wunused-result]
   176 |                 read(sd, response + 3, sizeof(response) - 4);
       |                 ^~~~~~
nagios.c: In function 'test_path_access':
nagios.c:122:17: warning: ignoring return value of 'asprintf' declared with attribute 'warn_unused_result' [-Wunused-result]
   122 |                 asprintf(&path, "%s/%s", p, program);
       |                 ^~~~~~
gcc -Wall -I.. -g -O2 -DHAVE_CONFIG_H -DNSCORE -c -o broker.o broker.c
gcc -Wall -I.. -g -O2 -DHAVE_CONFIG_H -DNSCORE -c -o nebmods.o nebmods.c
gcc -Wall -I.. -g -O2 -DHAVE_CONFIG_H -DNSCORE -c -o ../common/shared.o ../common/shared.c
gcc -Wall -I.. -g -O2 -DHAVE_CONFIG_H -DNSCORE -c -o query-handler.o query-handler.c
gcc -Wall -I.. -g -O2 -DHAVE_CONFIG_H -DNSCORE -c -o workers.o workers.c
workers.c: In function 'handle_worker_result':
workers.c:801:25: warning: ignoring return value of 'asprintf' declared with attribute 'warn_unused_result' [-Wunused-result]
   801 |                 asprintf(&error_reason, "timed out after %.2fs", tv_delta_f(&wpres.start, &wpres.stop)

```

Nous allons installer le groupe d'utilisateurs nécessaire pour accéder à la plateforme.

```

admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$
sudo make install-groups-users
groupadd -r nagios
useradd -g nagios nagios
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$

```

Nous allons accorder les autorisations requises.

```

admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$ sudo usermod -a -G nagios www-data
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$

```

Nous allons installer les binaires nagios avec la commande suivante.

```
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$ sudo make install
cd ./base && make install
make[1]: Entering directory '/home/admin-sys/Downloads/nagios-4.4.6/base'
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/bin
/usr/bin/install -c -s -m 774 -o nagios -g nagios nagios /usr/local/nagios/bin
/usr/bin/install -c -s -m 774 -o nagios -g nagios nagiosstats /usr/local/nagios/bin
make[1]: Leaving directory '/home/admin-sys/Downloads/nagios-4.4.6/base'
cd ./cgi && make install
make[1]: Entering directory '/home/admin-sys/Downloads/nagios-4.4.6/cgi'
make install-basic
make[2]: Entering directory '/home/admin-sys/Downloads/nagios-4.4.6/cgi'
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/sbin
for file in *.cgi; do \
    /usr/bin/install -c -s -m 775 -o nagios -g nagios $file /usr/local/nagios/sbin; \
done
make[2]: Leaving directory '/home/admin-sys/Downloads/nagios-4.4.6/cgi'
make[1]: Leaving directory '/home/admin-sys/Downloads/nagios-4.4.6/cgi'
cd ./html && make install
make[1]: Entering directory '/home/admin-sys/Downloads/nagios-4.4.6/html'
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/media
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/stylesheets
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/contexthelp
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/docs
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/docs/images
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/js
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/images
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/images/logos
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/includes
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/ssi
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/angularjs
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/share/angularjs/angular-1.3.9
```

Nous allons installer les scripts requis, d'abord le démon de démarrage :

```
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$ sudo make install-daemoninit
/usr/bin/install -c -m 755 -d -o root -g root /lib/systemd/system
/usr/bin/install -c -m 755 -o root -g root startup/default-service /lib/systemd/system/nagios.service
Created symlink /etc/systemd/system/multi-user.target.wants/nagios.service → /lib/systemd/system/nagios.service.

*** Init script installed ***

admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$
```

Maintenant le mode commande :

```
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$ sudo make install-commandmode
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/var/rw
chmod g+s /usr/local/nagios/var/rw

*** External command directory configured ***

admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$
```

Nous allons exécuter le script pour configurer.

```
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$ sudo make install-config
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/etc
/usr/bin/install -c -m 775 -o nagios -g nagios -d /usr/local/nagios/etc/objects
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/nagios.cfg /usr/local/nagios/etc/nagios.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/cgi.cfg /usr/local/nagios/etc/cgi.cfg
/usr/bin/install -c -b -m 660 -o nagios -g nagios sample-config/resource.cfg /usr/local/nagios/etc/resource.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/templates.cfg /usr/local/nagios/etc/objects/templates.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/commands.cfg /usr/local/nagios/etc/objects/commands.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/contacts.cfg /usr/local/nagios/etc/objects/contacts.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/timeperiods.cfg /usr/local/nagios/etc/objects/timeperiods.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/localhost.cfg /usr/local/nagios/etc/objects/localhost.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/windows.cfg /usr/local/nagios/etc/objects/windows.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/printer.cfg /usr/local/nagios/etc/objects/printer.cfg
/usr/bin/install -c -b -m 664 -o nagios -g nagios sample-config/template-object/switch.cfg /usr/local/nagios/etc/objects/switch.cfg

*** Config files installed ***

Remember, these are *SAMPLE* config files. You'll need to read
the documentation for more information on how to actually define
services, hosts, etc. to fit your particular needs.

admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$
```

Nous allons installer la configuration Apache pour Nagios.

```
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$ sudo make install-webconf
/usr/bin/install -c -m 644 sample-config/httpd.conf /etc/apache2/sites-enabled/nagios.conf
if [ 0 -eq 1 ]; then \
    ln -s /etc/apache2/sites-enabled/nagios.conf /etc/apache2/sites-enabled/nagios.conf; \
fi

*** Nagios/Apache conf file installed ***

admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$
```

Maintenant, nous allons activer le module cgi.

```
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$ sudo a2enmod rewrite cgi
Enabling module rewrite.
Enabling module cgi.
To activate the new configuration, you need to run:
    systemctl restart apache2
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$
```

Redémarrer le serveur apache.

```
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$ sudo systemctl restart apache2
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$
```

Nous allons créer l'utilisateur apache pour Nagios.

```
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$
sudo htpasswd -c /usr/local/nagios/etc/htpasswd.users nagiosadmin
New password:
Re-type new password:
Adding password for user nagiosadmin
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$
```

Nous allons ajouter Apache au pare-feu d'Ubuntu.

```
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$ sudo ufw allow apache
Rules updated
Rules updated (v6)
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$
```

Nous allons activer le pare-feu au démarrage du système.

```
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$ sudo ufw enable
Firewall is active and enabled on system startup
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$
```

Nous allons recharger les modifications du pare-feu.


```
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$ sudo ufw reload
Firewall reloaded
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$
```

Nous allons valider l'état du pare-feu.

```
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$ sudo ufw status
Status: active

To Action From
--
Apache ALLOW Anywhere
Apache (v6) ALLOW Anywhere (v6)

admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$
```

Nous allons installer les plugins Nagios avec la commande suivante.

```
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$ sudo apt install monitoring-plugins nagios-nrpe-plugin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libflashrom1 libftdi1-2 liblvm13 linux-headers-6.2.0-26-generic linux-hwe-6.2-headers-6.2.0-26
  linux-image-6.2.0-26-generic linux-modules-6.2.0-26-generic linux-modules-extra-6.2.0-26-generic
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libdbi1 libmysqlclient21 libpq5 libradcli4 liburiparser1 monitoring-plugins-basic monitoring-plugins-common
  monitoring-plugins-standard python3-gpg python3-samba python3-tdb rpcbind samba-common samba-common-bin
  samba-dsdb-modules smbclient
Suggested packages:
  icinga2 nagios-plugins-contrib fping postfix | sendmail-bin | exim4-daemon-heavy | exim4-daemon-light qstat
  heimdal-clients python3-markdown python3-dnspython cifs-utils
The following NEW packages will be installed:
  libdbi1 libmysqlclient21 libpq5 libradcli4 liburiparser1 monitoring-plugins monitoring-plugins-basic
  monitoring-plugins-common monitoring-plugins-standard nagios-nrpe-plugin python3-gpg python3-samba python3-tdb
  rpcbind samba-common samba-common-bin samba-dsdb-modules smbclient
0 upgraded, 18 newly installed, 0 to remove and 2 not upgraded.
Need to get 6880 kB of archives.
After this operation, 39,2 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://sn.archive.ubuntu.com/ubuntu jammy/main amd64 rpcbind amd64 1.2.6-2build1 [46,6 kB]
Get:2 http://sn.archive.ubuntu.com/ubuntu jammy-updates/main amd64 samba-common all 2:4.15.13+dfsg-0ubuntu1.5 [75,8 k
B]
Get:3 http://sn.archive.ubuntu.com/ubuntu jammy-updates/main amd64 smbclient amd64 2:4.15.13+dfsg-0ubuntu1.5 [473 kB]
Get:4 http://sn.archive.ubuntu.com/ubuntu jammy/main amd64 libdbi1 amd64 0.9.0-6build2 [26,6 kB]
Get:5 http://sn.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libmysqlclient21 amd64 8.0.34-0ubuntu0.22.04.1 [13
02 kB]
Get:6 http://sn.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libpq5 amd64 14.9-0ubuntu0.22.04.1 [141 kB]
Get:7 http://sn.archive.ubuntu.com/ubuntu jammy/universe amd64 libradcli4 amd64 1.2.11-1build1 [40,0 kB]
```

Nous allons accéder au répertoire Nagios et créer un nouveau répertoire :

```
cd /usr/local/nagios/etc
```

```
admin-sys@UbuntuServer:~/Downloads/nagios-4.4.6$
cd /usr/local/nagios/etc
admin-sys@UbuntuServer:/usr/local/nagios/etc$ sudo mkdir -p /usr/local/nagios/etc/servers
admin-sys@UbuntuServer:/usr/local/nagios/etc$
```

On accède au fichier de configuration avec un éditeur. Nous allons décommenter le répertoire /usr/local/nagios/etc/servers

```
GNU nano 6.2                                nagios.cfg *
# You can specify individual object config files as shown below:
cfg_file=/usr/local/nagios/etc/objects/commands.cfg
cfg_file=/usr/local/nagios/etc/objects/contacts.cfg
cfg_file=/usr/local/nagios/etc/objects/timeperiods.cfg
cfg_file=/usr/local/nagios/etc/objects/templates.cfg

# Definitions for monitoring the local (Linux) host
cfg_file=/usr/local/nagios/etc/objects/localhost.cfg

# Definitions for monitoring a Windows machine
#cfg_file=/usr/local/nagios/etc/objects/windows.cfg

# Definitions for monitoring a router/switch
#cfg_file=/usr/local/nagios/etc/objects/switch.cfg

# Definitions for monitoring a network printer
#cfg_file=/usr/local/nagios/etc/objects/printer.cfg

# You can also tell Nagios to process all config files (with a .cfg
# extension) in a particular directory by using the cfg_dir
# directive as shown below:

cfg_dir=/usr/local/nagios/etc/servers
#cfg_dir=/usr/local/nagios/etc/printers
#cfg_dir=/usr/local/nagios/etc/switches
#cfg_dir=/usr/local/nagios/etc/routers
```

Editer le fichier resource.cfg et appliquer les modifications

```
GNU nano 6.2                                resource.cfg *
#####
#
# RESOURCE.CFG - Sample Resource File for Nagios 4.4.6
#
#
# You can define $USERx$ macros in this file, which can in turn be used
# in command definitions in your host config file(s). $USERx$ macros are
# useful for storing sensitive information such as usernames, passwords,
# etc. They are also handy for specifying the path to plugins and
# event handlers - if you decide to move the plugins or event handlers to
# a different directory in the future, you can just update one or two
# $USERx$ macros, instead of modifying a lot of command definitions.
#
# The CGIs will not attempt to read the contents of resource files, so
# you can set restrictive permissions (600 or 660) on them.
#
# Nagios supports up to 256 $USERx$ macros ($USER1$ through $USER256$)
#
# Resource files may also be used to store configuration directives for
# external data sources like MySQL...
#
#####
# Sets $USER1$ to be the path to the plugins
$USER1$=/usr/local/nagios/plugins

# Sets $USER2$ to be the path to event handlers
#$USER2$=/usr/local/nagios/libexec/eventhandlers
```

Editer le fichier contacts.cfg

```
GNU nano 6.2                                objects/contacts.cfg

#####
#
# CONTACTS
#
#####

# Just one contact defined by default - the Nagios admin (that's you)
# This contact definition inherits a lot of default values from the
# 'generic-contact' template which is defined elsewhere.

define contact {
    contact_name      nagiosadmin          ; Short name of user
    use               generic-contact      ; Inherit default values from generic-contact template (defined elsewhere)
    alias             Nagios Admin        ; Full name of user
    email             nagios@localhost    ; <<***** CHANGE THIS TO YOUR EMAIL ADDRESS *****>>
}

#####
#
# CONTACT GROUPS
#
#####
```

Editer le fichier commands.cfg et appliquer les modifications à la fin.

```
GNU nano 6.2                                objects/commands.cfg *
#
# SAMPLE PERFORMANCE DATA COMMANDS
#
# These are sample performance data commands that can be used to send performance
# data output to two text files (one for hosts, another for services). If you
# plan on simply writing performance data out to a file, consider using the
# host_perfdata_file and service_perfdata_file options in the main config file.
#
#####

define command {
    command_name      process-host-perfdata
    command_line       /usr/bin/printf "%b" "$LASTHOSTCHECK$\t$HOSTNAME$\t$HOSTSTATES$\t$HOSTATTEMPTS$\t$HOSTSTATETYPES$\t$HOSTPERFDATA$\n"
}

define command {
    command_name      process-service-perfdata
    command_line       /usr/bin/printf "%b" "$LASTSERVICECHECK$\t$HOSTNAME$\t$SERVICEDESC$\t$SERVICESTATES$\t$SERVICEATTEMPTS$\t$SERVICESTATETYPES$\t$SERVICEPERFDATA$\n"
}

define command {
    command_name      check_nrpe
    command_line       $USER1$/check_nrpe -H $HOSTADDRESS$ -c $ARG1$
}


```

NRPE (Nagios Remote Plugin Executor) est un module avec lequel il est possible d'effectuer une surveillance active à l'aide d'un agent installé sur les ressources à surveiller. Après cela, nous démarrons et activons le service Nagios :

```
admin-sys@UbuntuServer:/usr/local/nagios/etc$ systemctl start nagios
admin-sys@UbuntuServer:/usr/local/nagios/etc$ systemctl enable nagios
admin-sys@UbuntuServer:/usr/local/nagios/etc$
```

Vérifier l'état du service Nagios


```

admin-sys@UbuntuServer:/usr/local/nagios/etc$ systemctl status nagios
● nagios.service - Nagios Core 4.4.6
   Loaded: loaded (/lib/systemd/system/nagios.service; enabled; vendor preset: enabled)
   Active: active (running) since Sun 2023-10-22 10:31:58 GMT; 55s ago
     Docs: https://www.nagios.org/documentation
   Main PID: 21706 (nagios)
    Tasks: 6 (limit: 2218)
   Memory: 2.8M
      CPU: 44ms
   CGroup: /system.slice/nagios.service
           └─21706 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg
             └─21707 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
               └─21708 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
                 └─21709 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
                   └─21710 /usr/local/nagios/bin/nagios --worker /usr/local/nagios/var/rw/nagios.qh
                     └─21711 /usr/local/nagios/bin/nagios -d /usr/local/nagios/etc/nagios.cfg

okt 22 10:31:58 UbuntuServer nagios[21706]: qh: echo service query handler registered
okt 22 10:31:58 UbuntuServer nagios[21706]: qh: help for the query handler registered
okt 22 10:31:58 UbuntuServer nagios[21706]: wproc: Successfully registered manager as @wproc with query handler
okt 22 10:31:58 UbuntuServer nagios[21706]: wproc: Registry request: name=Core Worker 21708;pid=21708
okt 22 10:31:58 UbuntuServer nagios[21706]: wproc: Registry request: name=Core Worker 21707;pid=21707
okt 22 10:31:58 UbuntuServer nagios[21706]: wproc: Registry request: name=Core Worker 21709;pid=21709
okt 22 10:31:58 UbuntuServer nagios[21706]: wproc: Registry request: name=Core Worker 21710;pid=21710
okt 22 10:31:58 UbuntuServer nagios[21706]: Successfully launched command file worker with pid 21711
okt 22 10:31:58 UbuntuServer nagios[21706]: HOST ALERT: localhost;DOWN;SOFT;1;(No output on stdout) stderr: execvp(/>
okt 22 10:32:35 UbuntuServer nagios[21706]: SERVICE ALERT: localhost;Current Load;CRITICAL;HARD;1;(No output on stdo
lines 1-26/26 (END)

```

Redémarrer le service apache

```

admin-sys@UbuntuServer:/usr/local/nagios/etc$ systemctl restart apache2
admin-sys@UbuntuServer:/usr/local/nagios/etc$

```

Aller au niveau du navigateur saisir :<http://@ipserveur/nagios>

Création des services pour afficher les statistiques DNS dans nagios.

```

GNU nano 6.2 /etc/check.sh
#!/bin/bash

dns_stats=$(dig @8.8.8.8 +short google.com)

if [ -n "$dns_stats" ]; then
    echo "DNS OK - $dns_stats"
    exit 0
else
    echo "DNS CRITICAL - Unable to resolve example.com"
    exit 2
fi

```

Editer le fichier commands.cfg et appliquer les modifications.

```
# host_perfdata_file and service_perfdata_file options in the main config file.
#
#####

define command {
    command_name    process-host-perfdata
    command_line    /usr/bin/printf "%b" "$LASTHOSTCHECK$\t$HOSTNAME$\t$HOSTSTATES$\t$HOSTATTEMPTS$\t$HOSTSTATETYPES$\t$HOSTPERFDATAFILE"
}

define command {
    command_name    process-service-perfdata
    command_line    /usr/bin/printf "%b" "$LASTSERVICECHECK$\t$HOSTNAME$\t$SERVICEDESC$\t$SERVICESTATES$\t$SERVICEATTEMPTS$\t$SERVICESTATETYPES$\t$SERVICEPERFDATAFILE"
}

define command{
    command_name check_nrpe
    command_line $USER1$/check_nrpe -H $HOSTADDRESS$ -c $ARG1$
}

define command{
    command_name check
    command_line /etc/check.sh
}
```

Editer le fichier machine.cfg

```
GNU nano 6.2 machine.cfg
define host {
    use                linux-server          ; Name of host template to use
    host_name          ubuntu                ; The name of the hostgroup
    alias              ubuntu                ; Long name of the group
    address             192.168.137.179
}
```

Exécuter le script

```
root@UbuntuServer:/etc# ./check.sh
DNS OK - ;; Invalid option

; <<>> DiG 9.18.12-0ubuntu0.22.04.3-Ubuntu <<>> @8.8.8.8 + short google.com
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 39863
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;short.                IN      A

;; AUTHORITY SECTION:
.                86169  IN      SOA      a.root-servers.net. nstld.verisign-grs.com. 2023102200 1800 900 60480
0 86400

;; Query time: 60 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Sun Oct 22 18:25:08 GMT 2023
;; MSG SIZE rcvd: 109

;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 25407
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 25407
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;google.com.          IN      A

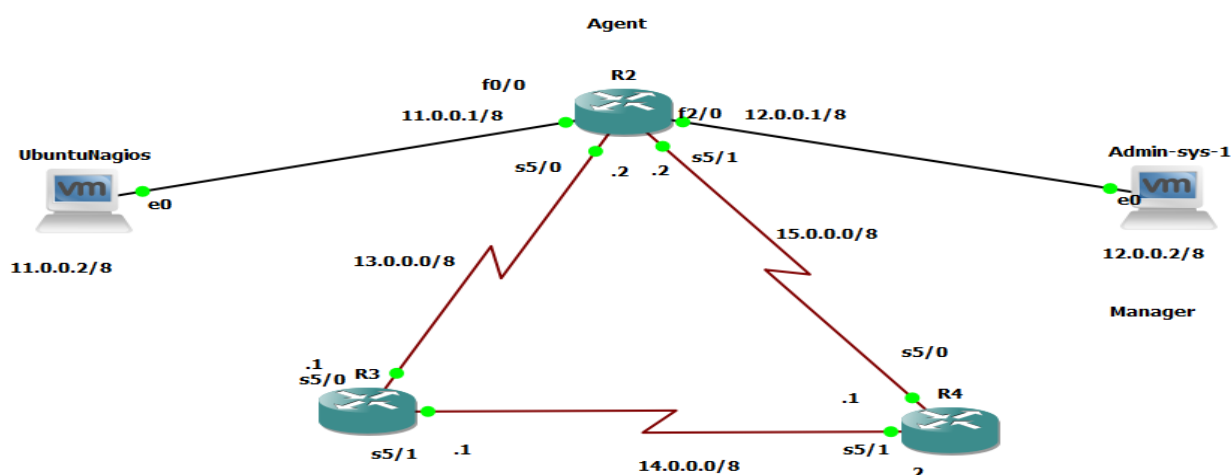
;; ANSWER SECTION:
google.com.          100     IN      A          142.250.185.14

;; Query time: 60 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Sun Oct 22 18:25:08 GMT 2023
;; MSG SIZE rcvd: 55
root@UbuntuServer:/etc#
```

Limit Results: 100						
Host	Service	Status	Last Check	Duration	Attempt	Status Information
localhost	Current Load	OK	10-22-2023 18:17:30	0d 6h 52m 47s	1/4	OK - load average: 0.78, 0.52, 0.47
	Current Users	OK	10-22-2023 18:19:21	0d 6h 52m 9s	1/4	USERS OK - 2 users currently logged in
	HTTP	OK	10-22-2023 18:20:39	0d 6h 51m 32s	1/4	HTTP OK: HTTP/1.1 200 OK - 10945 bytes in 0,020 second response time
	PING	OK	10-22-2023 18:18:07	0d 6h 50m 54s	1/4	PING OK - Packet loss = 0%, RTA = 0.21 ms
	Root Partition	WARNING	10-22-2023 18:19:58	0d 6h 50m 17s	4/4	DISK WARNING - free space: / 2925 MiB (15,84% inode=77%):
	SSH	CRITICAL	10-22-2023 18:20:39	0d 7h 4m 39s	4/4	connect to address 127.0.0.1 and port 22: Connection refused
	Swap Usage	OK	10-22-2023 18:18:44	0d 6h 54m 2s	1/4	SWAP OK - 69% free (1468 MB out of 2139 MB)
ubuntu	Total Processes	OK	10-22-2023 18:20:48	0d 6h 53m 24s	1/4	PROCS OK: 108 processes with STATE = RSZDT
	DNS Statistics	OK	10-22-2023 18:12:02	0d 0h 59m 59s	1/3	DNS OK - :: communications error to 12.0.0.2#53: timed out

Results 1 - 9 of 9 Matching Services

II. Topologie de l'activité 1 avec trois routeurs avec un agent, un serveur et un Ubuntu Desktop



Sur le répertoire /usr/local/nagios/etc/object, nous allons créer les fichiers machine.cfg et routeur.cfg

```

GNU nano 6.2                                     machine.cfg
define host {
    use                linux-server                ; Name of host template to use
    host_name          ubuntu                      ; The name of the hostgroup
    alias              ubuntu                      ; Long name of the group
    address            12.0.0.2
    parents            R1
}

```

```

GNU nano 6.2                                routeur.cfg
define host {
    use                linux-server          ; Name of host template to u>
    host_name          R1                    ; The name of the hostgroup
    alias              gateway              ; Long name of the group
    address            12.0.0.1
}

define host {
    use                linux-server          ; Name of host template to u>
    host_name          R2                    ; The name of the hostgroup
    alias              gateway              ; Long name of the group
    address            13.0.0.1
    parents            R1
}

define host {
    use                linux-server          ; Name of host template to u>
    host_name          R3                    ; The name of the hostgroup
    alias              gateway              ; Long name of the group
    address            15.0.0.1
    parents            R1
}

```

Sur le répertoire /usr/local/nagios/etc/

Editer le fichier nagios.cfg

```

GNU nano 6.2                                nagios.cfg
# LOG FILE
# This is the main log file where service and host events are logged
# for historical purposes.  This should be the first option specified
# in the config file!!!

log_file=/usr/local/nagios/var/nagios.log

# OBJECT CONFIGURATION FILE(S)
# These are the object configuration files in which you define hosts,
# host groups, contacts, contact groups, services, etc.
# You can split your object definitions across several config files
# if you wish (as shown below), or keep them all in a single config file.

# You can specify individual object config files as shown below:
cfg_file=/usr/local/nagios/etc/objects/commands.cfg
cfg_file=/usr/local/nagios/etc/objects/contacts.cfg
cfg_file=/usr/local/nagios/etc/objects/timeperiods.cfg
cfg_file=/usr/local/nagios/etc/objects/templates.cfg
cfg_file=/usr/local/nagios/etc/objects/machine.cfg
cfg_file=/usr/local/nagios/etc/objects/routeur.cfg

```

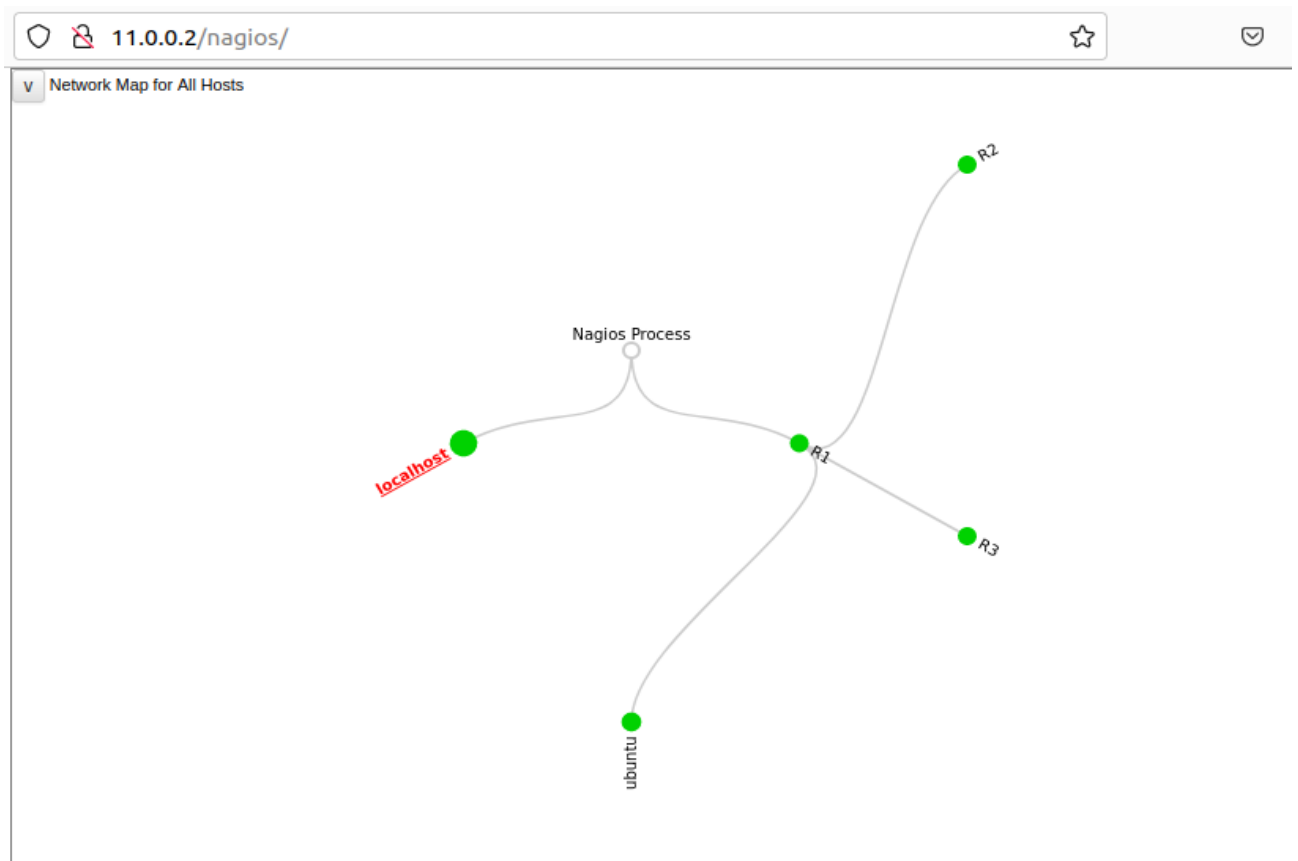
Redémarrer le service nagios

```

root@UbuntuServer:/usr/local/nagios/etc# systemctl restart nagios
root@UbuntuServer:/usr/local/nagios/etc#

```

Aller sur le navigateur saisir : <http://11.0.0.1/nagios>



III. Création des services pour surveiller si les interfaces des routeurs sont activées ou désactivées

Sur le répertoire `/usr/local/nagios/etc/object`, nous allons créer un fichier nommé `service.cfg` pour la surveillance des interfaces des routeurs et modifier le fichier `commands.cfg`

```
GNU nano 6.2 services.cfg
define service {
    use                generic-service
    host_name          R1
    service_description PING
    check_command       check_ping -H 11.0.0.1 -w 100.0,20% -c 500.0,60% -p 4 -t 6
}

define service {
    use                generic-service
    host_name          R2
    service_description PING
    check_command       check_ping -H 13.0.0.1 -w 100.0,20% -c 500.0,60% -p 4 -t 6
}

define service {
    use                generic-service
    host_name          R3
    service_description PING
    check_command       check_ping -H 15.0.0.1 -w 100.0,20% -c 500.0,60% -p 4 -t 6
}

define service {
    use                generic-service
    host_name          ubuntu
    service_description PING
    check_command       check_ping -H 12.0.0.2 -w 100.0,20% -c 500.0,60% -p 4 -t 6
}
```

```

root@UbuntuServer: /usr/local/nagios/etc/objects
GNU nano 6.2 commands.cfg
}

define command {
    command_name    check_ping
    command_line     $USER1$/check_ping -H $HOSTADDRESS$ -w $ARG1$ -c $ARG2$ -p $ARG3$ -t $ARG4$
}

```

Sur le répertoire /usr/local/nagios/plugins, nous allons choisir le plugin pour PING nommé 'check_ping' puis préciser les options suivantes :

-H pour donner l'adresse de l'interface du routeur,

-w 100.0,20% pour définir un seuil de 100ms

-c 500,0,60% pour définir un critique de 500ms

```

root@UbuntuServer:/usr/local/nagios/etc/objects# /usr/local/nagios/plugins/check_ping -H 11.0.0.1 -w 100.0,20% -c 500
.0,60% -p 4 -t 6
PING OK - Packet loss = 0%, RTA = 18.67 ms|rta=18.669001ms;100.000000;500.000000;0.000000 pl=0%;20;60;0
root@UbuntuServer:/usr/local/nagios/etc/objects# nano services.cfg
root@UbuntuServer:/usr/local/nagios/etc/objects# nano commands.cfg
root@UbuntuServer:/usr/local/nagios/etc/objects# /usr/local/nagios/plugins/check_ping -H 12.0.0.1 -w 100.0,20% -c 500
.0,60% -p 4 -t 6
PING OK - Packet loss = 0%, RTA = 8.76 ms|rta=8.761000ms;100.000000;500.000000;0.000000 pl=0%;20;60;0
root@UbuntuServer:/usr/local/nagios/etc/objects# systemctl restart nagios.service
root@UbuntuServer:/usr/local/nagios/etc/objects# /usr/local/nagios/plugins/check_ping -H 11.0.0.1 -w 100.0,20% -c 500
.0,60% -p 4 -t 6
PING OK - Packet loss = 0%, RTA = 19.70 ms|rta=19.697001ms;100.000000;500.000000;0.000000 pl=0%;20;60;0
root@UbuntuServer:/usr/local/nagios/etc/objects# /usr/local/nagios/plugins/check_ping -H 12.0.0.1 -w 100.0,20% -c 500
.0,60% -p 4 -t 6
PING OK - Packet loss = 0%, RTA = 8.44 ms|rta=8.438000ms;100.000000;500.000000;0.000000 pl=0%;20;60;0
root@UbuntuServer:/usr/local/nagios/etc/objects# /usr/local/nagios/plugins/check_ping -H 13.0.0.1 -w 100.0,20% -c 500
.0,60% -p 4 -t 6
PING OK - Packet loss = 0%, RTA = 44.74 ms|rta=44.743999ms;100.000000;500.000000;0.000000 pl=0%;20;60;0
root@UbuntuServer:/usr/local/nagios/etc/objects# /usr/local/nagios/plugins/check_ping -H 15.0.0.1 -w 100.0,20% -c 500
.0,60% -p 4 -t 6
PING OK - Packet loss = 0%, RTA = 38.77 ms|rta=38.766998ms;100.000000;500.000000;0.000000 pl=0%;20;60;0
root@UbuntuServer:/usr/local/nagios/etc/objects#

```

D'après l'image ci-dessus le nombre de packet of loss= 0%. Nous pouvons en déduire que la connectivité marche bien.

Host	Status	Last Check	Duration	Status Information
localhost	UP	10-22-2023 22:21:56	0d 10h 17m 49s	PING OK - Packet loss = 0%, RTA = 0.05 ms
R1	UP	10-22-2023 22:19:46	0d 0h 29m 46s	PING OK - Packet loss = 0%, RTA = 13.06 ms
ubuntu	UP	10-22-2023 22:19:42	0d 0h 24m 10s	PING OK - Packet loss = 0%, RTA = 23.44 ms
R2	UP	10-22-2023 22:19:06	0d 0h 29m 46s	PING OK - Packet loss = 0%, RTA = 49.36 ms
R3	UP	10-22-2023 22:19:06	0d 0h 29m 46s	PING OK - Packet loss = 0%, RTA = 47.56 ms

Results 1 - 5 of 5 Matching Hosts

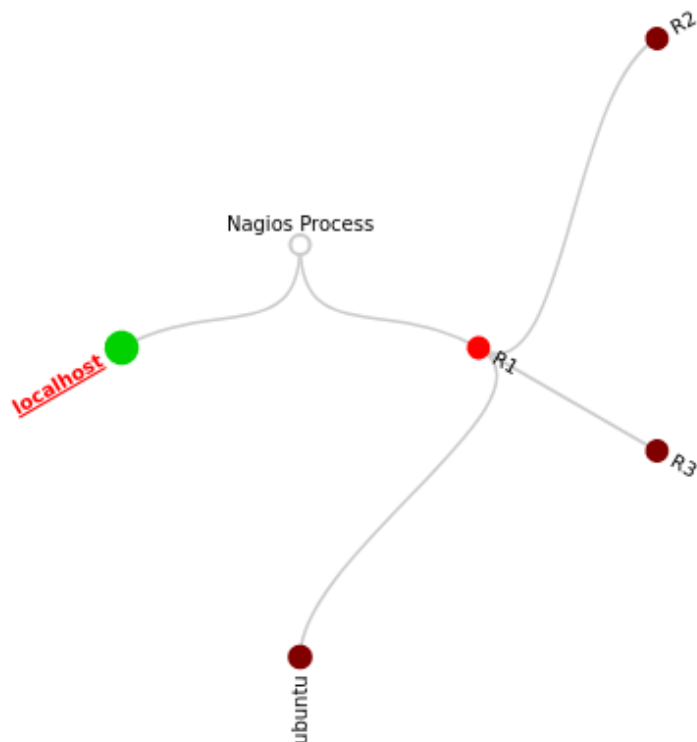
Maintenant, nous allons désactiver l'interface fastEthernet0/0 sur le routeur 2 pour tester la connectivité(PING)

```
R2(config)#interface fastEthernet 0/0
R2(config-if)#shut
R2(config-if)#shutdown
R2(config-if)#
*Oct 22 21:00:45.734: %LINK-5-CHANGED: Interface FastEthernet0/0, changed state to administratively down
*Oct 22 21:00:46.734: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to down
R2(config-if)#end
R2#w
*Oct 22 21:00:48.746: %SYS-5-CONFIG_I: Configured from console by console
R2#write
Building configuration...
[OK]
R2#
```

```
root@UbuntuServer:/usr/local/nagios/etc# /usr/local/nagios/plugins/check_ping -H 11.0.0.1 -w 100.0,20% -c 500.0,60%
p 4 -t 6
CRITICAL - Host Unreachable (11.0.0.1)
root@UbuntuServer:/usr/local/nagios/etc# /usr/local/nagios/plugins/check_ping -H 13.0.0.1 -w 100.0,20% -c 500.0,60%
p 4 -t 6
CRITICAL - Host Unreachable (13.0.0.1)
root@UbuntuServer:/usr/local/nagios/etc# /usr/local/nagios/plugins/check_ping -H 15.0.0.1 -w 100.0,20% -c 500.0,60%
p 4 -t 6
CRITICAL - Host Unreachable (15.0.0.1)
root@UbuntuServer:/usr/local/nagios/etc# /usr/local/nagios/plugins/check_ping -H 12.0.0.2 -w 100.0,20% -c 500.0,60%
p 4 -t 6
CRITICAL - Host Unreachable (12.0.0.2)
root@UbuntuServer:/usr/local/nagios/etc#
```

```
root@UbuntuServer:~# /usr/local/nagios/plugins/check_ping -H 11.0.0.1 -w 100.0,20% -c 500.0,60% -p 4 -t 6
PING CRITICAL - Packet loss = 100%|rta=500.000000ms;100.000000;500.000000;0.000000 pl=100%;20;60;0
root@UbuntuServer:~#
```

Après exécution de 4 paquets envoyés vers 11.0.0.1, nous remarquons que le nombre de packet of loss=100%. On peut dire déduire que le problème est du fait que l'interface du routeur int fa0/0 est 'DOWN'



Impossible d'accéder à 11.0.0.1

Host ▾	Status ▾	Last Check ▾	Duration ▾	Status Information
 R1   	DOWN	10-22-2023 22:39:55	0d 0h 6m 0s	CRITICAL - Host Unreachable (12.0.0.1)
R2   	UNREACHABLE	10-22-2023 22:40:03	0d 0h 5m 58s	CRITICAL - Host Unreachable (13.0.0.1)
R3   	UNREACHABLE	10-22-2023 22:40:03	0d 0h 5m 58s	CRITICAL - Host Unreachable (15.0.0.1)
localhost   	UP	10-22-2023 22:36:56	0d 10h 34m 43s	PING OK - Packet loss = 0%, RTA = 0.06 ms
ubuntu   	UNREACHABLE	10-22-2023 22:39:55	0d 0h 5m 1s	CRITICAL - Host Unreachable (12.0.0.2)

En résumé, Nagios est un outil de surveillance réseau essentiel pour les administrateurs système et les équipes informatiques, aidant à garantir la disponibilité et les performances des infrastructures informatiques.